Hafiz Ahmed Hassan Chohan

BS Computer Science | National University of Sciences and Technology (NUST)
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PROFESSIONAL PROFILE

I am a Machine Learning Engineer with a deep passion for research in Computer Vision (CV) and Deep Learning (DL). My research journey began with a strong curiosity about AI's ability to solve real-world problems, leading me to work on various impactful projects. As a Research Assistant at PakThai Lab, I collaborated on developing a real-time traffic violation detection system, which fueled my interest in applying AI to smart cities. Additionally, I contributed to projects like a web-based medical data annotation platform, aimed at automating fracture detection in X-rays, and a landmark detection system for autonomous drones. I am particularly driven by the potential of AI to transform industries, and I aim to continue pushing the boundaries of AI research, while working towards publishing my first research paper. My focus is on innovative projects that bridge theory with real-world applications in autonomous systems, healthcare, and urban technologies.

EDUCATION

- BS in Computer Science School of Electrical Engineering and Computer Science , 3.62
- Computer Science SEECS , 3.62 (2025)

WORK & INTERNSHIP EXPERIENCE

Signal and Machine Learning Lab
08-May-2023 - 28-Jul-2023

InLights
04-Mar-2024 - 24-May-2024

PakThai Lab NUST
01-May-2024 - 31-Mar-2025

Walee Technologies
13-May-2024 - 27-Apr-2025

PROJECTS

Gesture-Controlled Snake Game

- Developed a gesture-controlled snake game using OpenCV and pygame, offering a hands-free gaming experience through real-time hand gesture detection.
- Integrated cvzone for real-time hand tracking and gesture recognition, providing responsive and intuitive gameplay.

Convert to EBooks

- Created a system to convert physical book images or PDFs into fully formatted eBooks using deep learning and OCR technologies.
- Trained a deep neural network for segmentation, accurately extracting text from diverse fonts and layouts.
- Follow-up work based on the paper: Shafait, F., & Smith, R. (2010).

ANPR– Automatic Number Plate Recognition System

- Developed a real-time ANPR system using computer vision and deep learning, accurately detecting and recognizing vehicle license plates.

- Built a robust image processing pipeline using OpenCV and integrated it with easy-ocr for character recognition, achieving reliable performance under various lighting an